

PHYS 3200 - Fall 2026

End of Topic Quiz 1

September 8, 2025

Name and ID #:

Instructions:

1. You may use:
 - A calculator.
 - Notes.
 - Writing utensil: pencil and eraser are best.
2. If you have a question about a problem (confused about the situation, need some missing piece of information, etc.), please raise your hand and ask!
3.

 for each part.
4. Credit will not be given if your answers are too messy or obscure to read. Remember - a grader who isn't squinting and spending extra time trying to decipher mysterious scratches is a happy grader.
5. This quiz must be turned in by the end of the class period.

1. A particle of mass m moves in the xy -plane. Initially, $t = 0$, its position and velocity are

$$\vec{r}(0) = x_0\hat{\mathbf{x}}, \quad \vec{v}(0) = v_0\hat{\mathbf{y}}$$

For $t > 0$, it is acted upon by a constant force, where F_0 is a positive constant.

$$\vec{F} = F_0(2\hat{\mathbf{x}} - 1\hat{\mathbf{y}})$$

- (a.) Use Newton's second Law to find the acceleration $\vec{a}(t)$ or $\ddot{\vec{r}}(t)$.

- (b.) Find the velocity $\vec{v}(t)$ or $\dot{\vec{r}}(t)$.

- (c.) Find the position $\vec{r}(t)$.

- (d.) At what time is the y -component of the particle's velocity equal to zero?

2. A particle of mass m moves in a circle of **constant radius** R . Its angular position is:

$$\phi(t) = \omega_0 t + \frac{1}{2} \alpha t^2$$

where ω_0 and α are constants.

- (a.) Determine $\dot{\mathbf{r}}$, $\ddot{\mathbf{r}}$, $\dot{\phi}$ and $\ddot{\phi}$.

- (b.) Using the polar-coordinate expression for acceleration:

$$\vec{a} = (\ddot{r} - r\dot{\phi}^2)\hat{\mathbf{r}} + (r\ddot{\phi} + 2\dot{r}\dot{\phi})\hat{\phi}$$

find $\vec{a}(t)$.

- (c.) Use Newton's second law to find the radial and tangential components of the net force.

- (d.) Briefly comment on the physical role of each force component.